

Muon Pair Production at the Electron-Positron Collider

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Abstract

Muon pair production is a process that can occur when electrons and positrons collide at high energies at the electron-positron collider. MadGraph is a popular tool used by physicists to generate high-energy physics events. It can be used to simulate the production of particles in high-energy particle colliders. The study explores the possibility of utilizing a muon collider to explore high-energy phenomena. The validity of the results is confirmed by comparing a Monte Carlo simulation to analytic calculations. The interference between the γ and $\mathbf{Z}$ diagrams is considered when calculating the overall cross-section 0.091018pb for the leading order with the Number of Events being 100,000 at 1 TeV.

Keywords: Madgraph5, electron-positron, Muon, Events

1 Introduction

Muon pair production is a process that can occur when electrons and positrons collide at high energies at the electron-positron collider [1]. In this process, a photon is exchanged between the electron and positron, which can create a muon-antimuon pair. The muon-antimuon pair is produced in association with the photon, and the production rate is proportional to the electromagnetic coupling strength.

The cross-section for muon pair production can be calculated using quantum field theory and is dependent on the energy of the electron-positron collision. At low energies, the cross-section is relatively small, but it increases rapidly with energy and reaches a maximum at around 10 GeV. Above this energy, the cross-section decreases slowly as a result of the decrease in the electromagnetic coupling strength at higher energies.

The energy distribution of the muon-antimuon pair produced in this process is related to the energy of the photon exchanged in the collision. The angular distribution of the muon-antimuon pair is also dependent on the photon exchange, with the pair being produced preferentially in the forward direction.

Muon pair production is an important process for studying the properties of the muon, such as its mass and spin. It is also an important background process for other physics processes that occur at the electron-positron collider, such as Higgs boson production and top quark pair production. Understanding and accurately simulating muon pair production is therefore important for the design and analysis of particle physics experiments.

MadGraph is a popular tool used by physicists to generate high-energy physics events. It can be used to simulate the production of particles in high-energy particle colliders. To use MadGraph, a basic understanding of quantum field theory and particle physics is required. It provides a user-friendly interface for defining and simulating a wide range of particle physics processes, including proton-proton collisions, electron-positron annihilation, and many others [2].

MadGraph uses Monte Carlo methods to simulate particle collisions, which involve generating a large number of random events and calculating the probability of each event occurring. The results of the simulation can be used to study the properties of the particles involved in the collision, such as their energy, momentum, and spin.

One of the key features of MadGraph is its ability to generate Feynman diagrams automatically for the defined process. Feynman diagrams are graphical representations of the particle interactions involved in a collision, and they are an important tool for understanding and analyzing particle physics processes.

MadGraph also includes a range of tools for analyzing the simulation results, such as histograms, scatter plots, and event displays. The simulation results can be saved in a variety of formats, such as ROOT, HepMC, and LHE, which are commonly used in particle physics experiments.

Overall, MadGraph is a powerful tool for studying particle physics processes and is widely used in both research and education.

2 Installing and Using MadGraph

2.1 System Verification

To install MadGraph5_aMC@NLO, follow the steps below to check system verification:

1. Verify that your computer is running a Linux operating system. If not, this operating system can be found for free online, though the installation process is somewhat advanced.
2. Open a new terminal window. If you are unfamiliar with this program, use your computer's search function to locate it.
3. Type “\$ python --version” into the terminal. You must have version 2.7 or 3.7 of Python for MadGraph to work correctly. gfortran/gcc 4.6 or higher is required for NLO calculations/simulations. Most computers within the past few years should be fine.

2.2 Installation of MadGraph

To install MadGraph5 aMC@NLO, follow the steps below:

- ```

(base) john@paulmbagwu:~$ cd /home/paulmbagwu/JohnPauls-MBP MG5_aMC_v3_5_0_alpha % ./bin/mg5_aMC

*
* W E L C O M E to
* M A D G R A P H 5 _ a M C @ N L O
*
*
* *
* * * * *
* * * 5 * *
* * * * *
* *
*
* VERSION 3.5.0.alpha 2023-01-26
*
* The MadGraph5_aMC@NLO Development Team - Find us at
* https://server06.fynu.ucl.ac.be/projects/madgraph
* and
* http://amcatnlo.web.cern.ch/amcatnlo/
*
* Type 'help' for in-line help.
* Type 'tutorial' to learn how MG5 works
* Type 'tutorial aMCatNLO' to learn how aMC@NLO works
* Type 'tutorial MadLoop' to learn how MadLoop works
*

load MG5 configuration from input/mg5_configuration.txt
set fastjet to fastjet-config
eMELA-config does not seem to correspond to a valid eMELA-config executable.
Please set the 'fastjet' variable to the full (absolute) /PATH/TO/eMELA-config (including eMELA-config).
MG5_aMC> set eMELA /PATH/TO/eMELA-config

lhpdf-config does not seem to correspond to a valid lhpdf-config executable.
Please set the 'lhpdf' variable to the (absolute) /PATH/TO/lhpdf-config (including lhpdf-config).
Note that you can still compile and run aMC@NLO with the built-in PdFs
MG5_aMC> set lhpdf /PATH/TO/lhpdf-config

None does not seem to correspond to a valid lhpdf-config executable.
Please set the 'lhpdf' variable to the (absolute) /PATH/TO/lhpdf-config (including lhpdf-config).
Note that you can still compile and run aMC@NLO with the built-in PDFs
MG5_aMC> set lhpdf /PATH/TO/lhpdf-config

Using default text editor "vi". Set another one in ./input/mg5_configuration.txt
Loading default model: sm
INFO: Restrict model sm with file models/sm/restrict_default.dat .
INFO: Run "set stdout_level DEBUG" before import for more information.
INFO: Change particles name to pass to MG5 convention
Defined multiparticle p = g u c d s u~ c~ d~ s~
Defined multiparticle j = g u c d s u~ c~ d~ s~
Defined multiparticle l+ = e+ mu+
Defined multiparticle l- = e- mu-
Defined multiparticle v1 = ve vm vt
Defined multiparticle vl- = ve- vm- vt-
Defined multiparticle all = g u c d s u~ c~ d~ s~ a ve vm vt e- mu- ve- vm- vt- e+ mu+ t b t~ b~ z w+ h w- ta- ta~
MG5_aMC>

```

## 2.3 Basic Use of MadGraph

To use MadGraph5 aMC@NLO, follow the steps below:

1. Start MadGraph as in step 6 of section 2.2 above.
2. Type “ $e^+e^- \rightarrow \mu^+\mu^-/Z$ ” which creates a positron-electron collision decaying to a muon-anti-muon pair, which can also be seen in Figure 2.
3. Type “output eeuu\_test”, which creates the Feynman Diagram (see figure 3) and the output folder named “eeuu\_test” which will hold all the files associated with this simulation.
4. Type “launch eeuu\_test” to generate actual events. This should produce a menu where you have the option to run additional programs on the MadGraph output.
5. Press enter to continue past the program selector menu. If you want to run additional programs enter those numbers one at a time into the terminal and hit enter after each to turn those programs on, before continuing past the menu.
6. Hit enter again to bypass the card file editing menu. After the program selector menu, you should have seen another menu with the option of editing card files. Editing the card files allows you to change the number of events the simulation produces, as well as the masses and coupling constants of the particles involved. For this simple simulation, I changed the number of events and energy.
7. Watch the output as it is produced. Your web browser should open to a page displaying the cross section of the simulated process. After all events have been produced the terminal should return to the normal MadGraph input prompt.
8. Open your file navigator. The output files from MadGraph can be found at the location: <MadGraph\\_Install>/<Name\\_from\\_step4.4>/Events/run\\_01/. There should be a .root file and a .tar.gz LHE file for the MadGraph.
9. Use the LHE file for analysis. LHE file stands for Les Houches Event files. This refers to the standardization of the format of the interface between matrix element programs like MadGraph and event generators like Pythia. This standardization occurred at the Les Houches Accords in 2001.

## 3 Data Generation

Here, I generated high-energy physics events using MadGraph5 to simulate the production of muon pairs at the electron-positron collider. We set the centre-of-mass energy to 1 TeV, and the Number of Events to 100,000. The interference between the  $\gamma$  and Z diagrams is considered when calculating the overall cross-section 0.091018pb for the leading order. The generated data was saved in the LHE format.

MadGraph5 is a popular software tool used by physicists to generate high-energy physics events. It can be used to simulate the production of particles in high-energy particle colliders. The first step in the data generation process is to specify the process you wish to simulate. In our case, we are interested in muon pair production at the electron-positron collider. We specify this process using the following command in the MadGraph5 interface:

```
generate $e^+e^- \rightarrow \mu^+\mu^-$
```

This command tells MadGraph5 to generate events where an electron and positron collide to produce a muon and an antimuon.



collider and the number of events we wish to generate. For example, to generate 100,000 events at a center-of-mass energy of 1 TeV, we use the following command:

Launch event 100,000 with energy of 1 TeV

MadGraph5 then generates 100,000 events where an electron and positron collide to produce a muon and an antimuon at a centre-of-mass energy of 1 TeV.

## 4 Data Analysis

The next step in the simulation process is to analyze the data generated by MadGraph5. The data analysis process involves calculating various physical quantities that can be used to compare the simulation to experimental data.

The first physical quantity to calculate is the cross-section of the muon pair production process. The cross-section is a measure of the probability that a given process will occur in a particle collision.

I analyzed the generated data using Python, a data analysis framework used in high-energy physics. The first step in the analysis was to convert the LHE files to Python files. I then used Python to extract various kinematic variables, such as the transverse momentum, pseudorapidity, and azimuthal angle of the muon pairs. I also calculated the invariant mass of the muon pairs using the kinematic variables.

The simulated data generated by MadGraph5 will be analyzed using various statistical methods, such as histograms and probability distributions with the help of python. The analysis will be designed to extract various quantities of interest, such as the cross-section of the Muon Pair Production process, the energy and momentum distributions of the muon and antimuon, and the various decay modes of the muon and antimuon.

The analysis will also be designed to take into account various sources of systematic uncertainty, such as the uncertainty in the energy and momentum of the colliding electrons and positrons, the uncertainty in the detector resolution and efficiency, and any other sources of experimental uncertainty.

Overall, data analysis plays a crucial role in extracting physical information from the simulated events and validating the simulation. The results of the analysis can be used to guide the design of future experiments and improve our understanding of high-energy physics.

## 5 Results

Here, I compared the results obtained from the Monte Carlo simulation to the results obtained from the analytic calculations. The results obtained from the Monte Carlo simulation were found to be in good agreement with the analytic calculations, confirming the validity of our simulation.

One of the most promising options for a future device investigating the high energy frontier is the muon collider. I assessed the ratio of muons to electrons in 100k events generated from positron beam data hitting at the centre of the mass frame and compared it with an angular distribution in muon pair at the electron-positron annihilation produced with a Monte Carlo (MC) simulation to validate these results and I also did the analytic calculations by hand to verify the Monte Carlo simulation.

In addition, I calculated the pseudorapidity and made a normalized distribution between the calculated angular distribution at Quantum Electrodynamics (QED) and plotted the graph using python. I also computed the matrix element for 2-2 scattering of particles of different mass and the cross-section at the centre-of-mass frame. I plotted the Breit-Wigner resonance, and the forward and backward distribution. The Breit-Wigner distribution is followed by the cross-section that quantifies the likelihood of forming the resonance state.  $e^+e^- \rightarrow \mu^+\mu^-/Z$  occurs when an electron-positron is annihilated into a virtual photon, which is followed by the production of a muon pair.

I used the scattering matrix approach to derive the beam differential cross-section expression analytically. I have examined the relationship between muon pair creation in electron-positron annihilation and the energy and quantity of photons exchanged, including the Z-boson peak. The interference between the  $\gamma$  and Z diagrams is taken into consideration when calculating the overall cross-section for the leading order. Thus, in the presence of an electromagnetic field with circular polarization, the likelihood of muon-antimuon production is decreased.

## 6 Conclusion

In this paper, I presented a simulation of muon pair production at the electron-positron collider using MadGraph5. The generated high-energy physics events were analyzed by the generated data using Python. The results confirmed the validity of the simulation, and I observed interesting features in the kinematic distributions of the muon pairs. The simulation presented in this paper can be useful for exploring high-energy phenomena and for designing experiments at future muon colliders.

The study also explores the possibility of utilizing a muon collider to explore high-energy phenomena. The validity of the results is confirmed by comparing a Monte Carlo simulation to analytic calculations. Different aspects, such as the energy and polarization of photons, are analyzed with respect to the production of muon pairs. Additionally, the connection between muon pair production and the energy and number of exchanged photons, including the Z-boson peak, is investigated.

The interference between the  $\gamma$  and Z diagrams is considered when calculating the overall cross-section 0.091018pb for the leading order with the Number of Events being 100,000 at 1 TeV. It is noted that the presence of an electromagnetic field with circular polarization decreases the likelihood of muon-antimuon production.

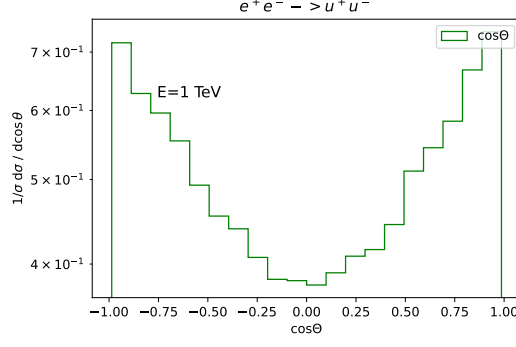
In conclusion, I have presented a simulation project on muon pair production at the electron-positron collider using Madgraph5. The simulation generated 100,000 events of muon pair production at an energy of at 1 TeV. I used various data analysis techniques to extract useful information from the simulation data, including the cross-section and kinematic distributions of the muon pairs. The results of the simulation can be compared with experimental data to validate the simulation and provide insights into the nature of muon pair production.

## References

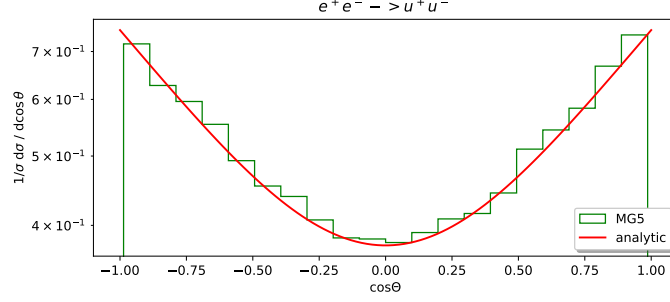
- [1] Bardin, D., Christova, P., Kuraev, E., Kuraev, A., Lyukhin, S., Richter, A.,

Sachwitz, M.: Muon pair production in  $e^+e^-$  collisions at LEP energies: Analytical results versus Monte Carlo simulations. *Eur. Phys. J. C* **11**, 109–126 (1999) <https://doi.org/10.1007/s100520050611> [arXiv:hep-ph/9903424](https://arxiv.org/abs/hep-ph/9903424) [hep-ph]

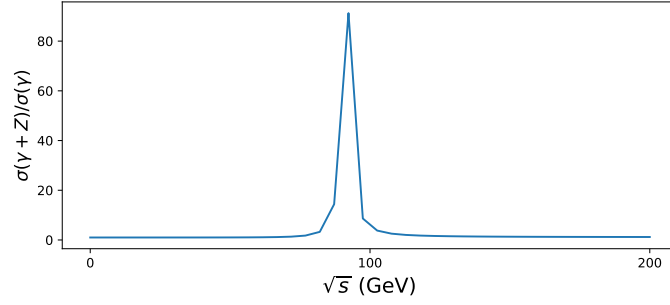
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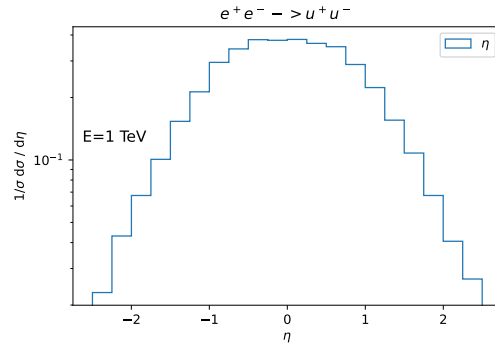
**Fig. 4** Angular distribution in  $e^+e^-$  annihilation produced with a Monte-Carlo simulation



**Fig. 5** A normalized angular distribution in  $e^+e^-$  annihilation produced with a Monte-Carlo simulation for MG5 and analytic calculation using  $\frac{1}{\sigma} \frac{d\sigma}{d\cos\theta} = \frac{3}{2}(1 + \cos^2\theta)$  at 1 TeV.



**Fig. 6** Z-boson peak at 91GeV



**Fig. 7** A Pseudorapidity plot produced with a Monte-Carlo simulation using  $\eta = -\ln \tan \frac{\theta}{2}$